

Hand Gesture Scrolling Apps

is there any application or software that can detect my hand gesture through my laptop and and basically like scroll the mic, so i can read easily ?

Yes. You can use your laptop webcam to detect hand gestures and translate them into mouse scrolling, so you can read papers/webpages without touching the mouse.

Given that you're on Linux/Hyprland, I'd look at these first:

- **KineMouse** — probably the best fit for your setup. It supports Windows, macOS, and Linux, and specifically handles **Wayland** using **uinput**. It uses MediaPipe hand tracking and provides cursor control; its architecture is fairly clean if you later want to customize gestures. [GitHub](#)
- **gesture-mouse** — simpler MediaPipe-based virtual mouse. It explicitly supports **scrolling**, with a two-finger gesture controlling scroll, and claims Linux/Windows/macOS support. [GitHub](#)
- **AI Virtual Mouse** — another straightforward Python implementation. One gesture enters scrolling mode, then moving the hand vertically scrolls the active window. [GitHub](#)
- **Swoosh** — more like a polished gesture-control layer rather than a small Python demo. It runs at the OS level and supports Windows/macOS/Linux. [GitHub](#)

For what you're describing, I wouldn't actually implement a full "virtual mouse." I'd make the interaction much simpler:

Webcam → detect hand → recognize only 2–3 gestures → emit scroll events

For example:

```

👉 Open palm
  ↓
activate reading mode
👉 Move hand upward
  ↓
scroll page down slowly
👉 Move hand downward
  ↓
scroll page up
👊 Fist
  ↓
pause / stop scrolling
  
```

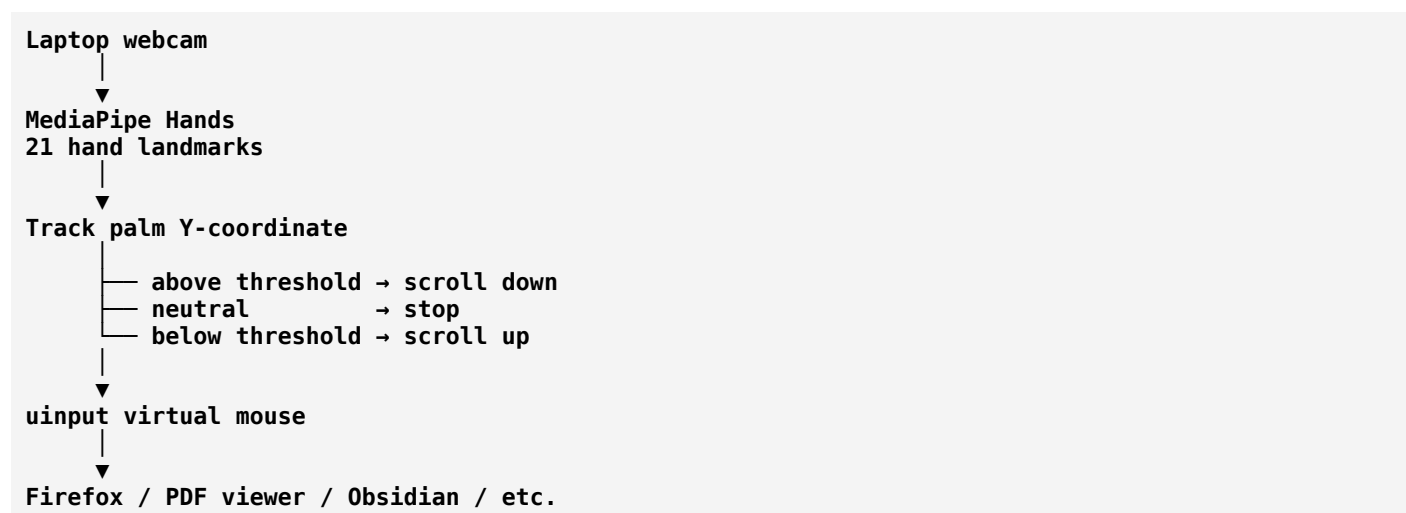
Or an even better reading-oriented design:

```

Palm held high
  ↑
faster scroll
—— neutral zone —— ← no scrolling
  ↓
slower/reverse
Palm held low
  
```

That would behave almost like a **gesture-controlled scroll wheel**. The farther your hand moves from the neutral position, the faster the document scrolls. You wouldn't need to repeatedly wave your hand.

Technically this is quite easy with **MediaPipe Hand Landmarker + OpenCV + Linux uinput/evdev**:



Since you're using **Hyprrland/Wayland**, **pyautogui**/**pynput** can be unreliable because Wayland restricts synthetic input. Something using **uinput**, such as the Linux implementation described by KineMouse, is the cleaner approach. [GitHub](#)

For your specific goal—**sitting back and reading research papers**—I'd actually make a tiny dedicated utility instead of installing a huge gesture-mouse project. It could run in the tray, use almost no UI, and have only:

open palm = activate → move palm up/down = scroll → fist = pause.

That would probably feel considerably better than trying to control the entire cursor with your hand.